

SIGHT
Senses

SIGHT Senses : feel the Way, hear the World.

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PRESENTATION

More than 2.2 billion
visually impaired
people in the world

Numerous challenges for
them in daily life.

Project goal : Create a wearable device
to help visually impaired individuals to
detect and avoid obstacles using
sensors and real-time feedback.



ALGORITHMS

Programming
Language: Python =
easier

1 IMAGE PROCESSING FUNCTIONS

library used : OpenCV
Different functions which detect lines, squares,
crosses
thanks to the camera. A function which enables to
detect distance. Parameters associated to
detection are returned

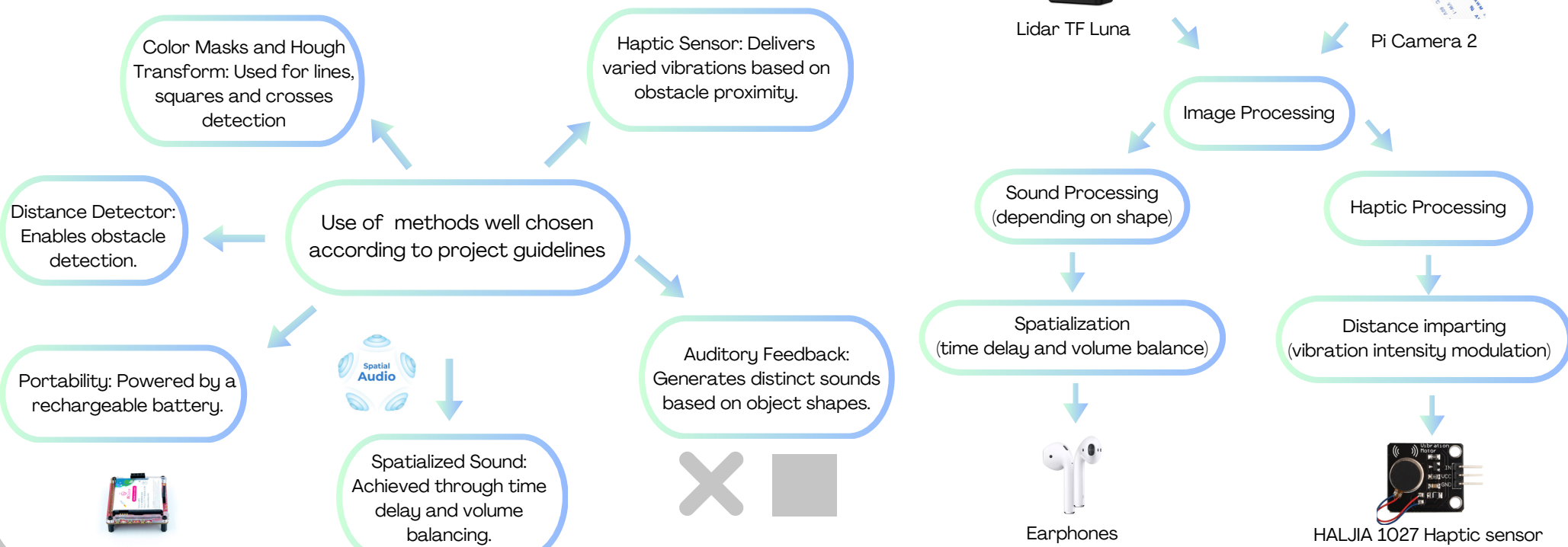
2 SOUND PROCESSING FUNCTIONS

library used : Pydub
function : depending on the type of
object and an axe x, play a sound in a
direction

3 HAPTIC FUNCTIONS

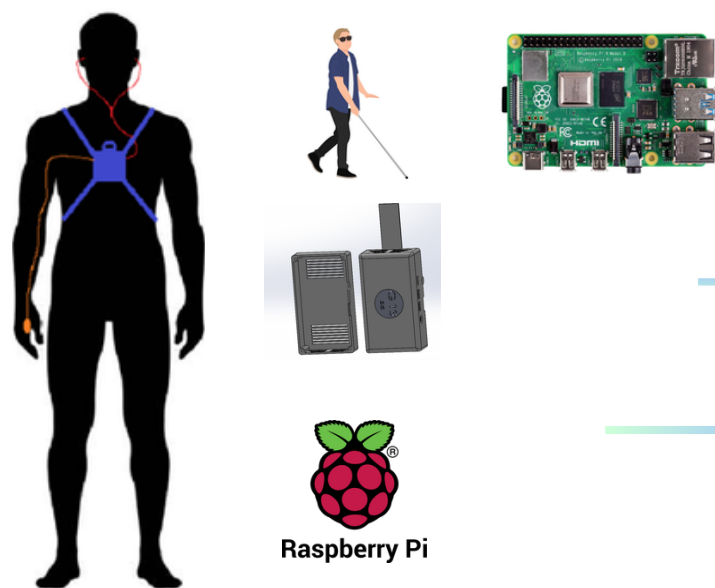
library used : RPi.GPIO
one fuction providing a vibration
depending on intensity and time
another one providing continous or
jerky vibration depending on the
distance from an obstacle

HOW DOES IT WORK ?



DEVICE

TESTS

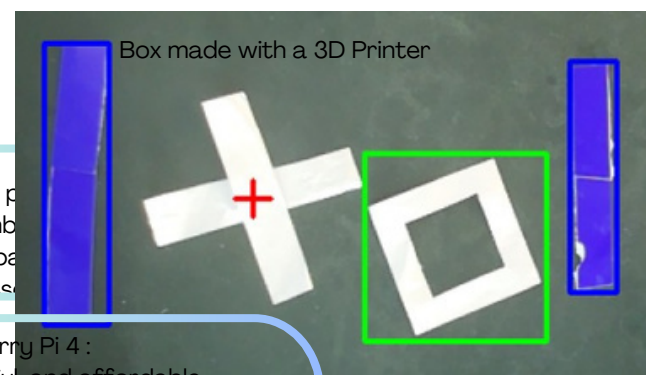


Size : 9.0 x 6.2

A 3D design reflecting the p
ecological commitment, comb
model that optimizes the ba
minimize thermal loss



Raspberry Pi 4 :
compact, powerful, and affordable
single-board computer ideal for DIY,
IoT, and educational projects



Shapes detected by the camera
Obstacles detected with the lidar tf luna too